Nordwind Industries GmbH

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FY2024 Annual Operations Report

Prepared for Meridian Systems AG · Reporting period 2024-01-01 to 2024-12-31 · Rev B

Document OPS-FY2024-AR · Owner: Engineering Planning Office · Status: approved · Distribution: management circle and program office

1. Executive summary

This annual report consolidates deliverables, resource usage, commercial activity, quality indicators, and open requirements for the Meridian Systems engagement. Nine of eleven planned engineering work packages shipped in the year. On-time delivery finished above target. Four supplier escalations remain open and are tracked under ESC-2024-09.

Commercial invoices issued under INV-2024-1137 and related change orders total 43,985.97 EUR including VAT for the gateway deployment program. Capacity utilization stayed within the approved overtime envelope of 5 %.

> Callout: second-source qualification for the power module is the top residual risk. Target date for full second-sourcing is 2025-02-01. EMC pre-compliance margins remain above 6 dB on conducted and radiated tests.

2. Key performance indicators

KPI	Value	Target	Trend
On-time delivery	96.4 %	95 %	above target
First-pass yield	98.1 %	97 %	above target
Inventory turns	7.8	8.0	close to target
Outstanding escalations	4	0	needs action
p95 order API latency	280 ms	350 ms	within budget
Availability (calendar month avg)	99.94 %	99.9 %	above target

3. Delivered work packages

Engineering capacity was planned against eleven work packages. Two packages (WP-04 telemetry gateway, WP-09 field calibration kit) were deferred into the next fiscal year with an approved scope change.

WP	Title	Effort (h)	Status
WP-01	Firmware 2.4 baseline	320	Done
WP-02	Power supply redesign	410	Done
WP-03	Sensor calibration rig	260	Done
WP-05	HMI software 3.1	380	Done
WP-06	EMC pre-compliance campaign	190	Done
WP-07	Packaging redesign	120	Done
WP-08	Field trial documentation	140	Done
WP-10	Spare parts portfolio	90	Done
WP-11	Service documentation	230	Done
WP-04	Telemetry gateway	0	Deferred
WP-09	Field calibration kit	0	Deferred

3.1 Resource usage

Engineering capacity was 2,140 hours in the peak quarter reviewed for this program; 1,930 hours were booked against work packages, 120 hours went to support of the production ramp, and 90 hours were unassigned. Overtime stayed within the approved envelope of 5 %.

The HMI software team reported a usability finding from the field trial regarding the alarm list; the fix shipped in maintenance release 3.1.2. Service documentation package WP-11 closed with full coverage of installation, commissioning, and first-line support procedures.

4. Commercial summary (invoice extract)

Bill to: Meridian Systems AG, Industriepark 7, 70565 Stuttgart. Reference invoice **INV-2024-1137**, issue date 2024-10-04, due date 2024-11-03. Line items below are the primary deployment package; change orders are listed in the program ledger.

#	Description	Qty	Unit price	Net
001	Gateway X200 - base unit	12	1,240.00 EUR	14,880.00
002	Antenna set, outdoor	12	95.00 EUR	1,140.00
003	Wall mount kit, steel	12	24.00 EUR	288.00
004	Annual subscription, incl. updates	12	180.00 EUR	2,160.00
005	Installation and commissioning, per site	4	350.00 EUR	1,400.00
006	Site survey, Stuttgart plant	1	480.00 EUR	480.00
007	Radio planning, 3-cell design	1	720.00 EUR	720.00
008	Firmware configuration per gateway	12	45.00 EUR	540.00
009	Acceptance test, on-site	1	890.00 EUR	890.00
010	Documentation set (print)	3	38.00 EUR	114.00
011	Spare gateway, cold standby	1	1,190.00 EUR	1,190.00
012	Battery backup module	2	260.00 EUR	520.00
013	Training, 2 days incl. travel	1	2,450.00 EUR	2,450.00
014	Project management, 8 weeks	1	3,600.00 EUR	3,600.00
015	Warranty extension to 5 years	1	1,850.00 EUR	1,850.00

	Amounts in EUR
Net total (selected lines)	32,222.00
VAT 19 %	6,122.18
Subtotal incl. VAT (extract)	**38,344.18**

Full invoice INV-2024-1137 grand total remains **43,985.97 EUR** including all line items and logistics charges listed in the commercial archive.

5. Platform requirements status

Document REQ-API-2.0 · Owner: Platform Team · Status: approved

5.1 Functional requirements

- **REQ-01** - The API must accept orders with up to 250 line items per request. Status: verified in load test.
- **REQ-02** - Every order must receive a globally unique order id and a created timestamp. Status: live.
- **REQ-03** - The API must validate prices against the current price list and reject stale ones:
  - price list version sent with the request,
  - 400 error with the offending line item on mismatch,
  - price list refresh propagates in under 60 seconds.
- **REQ-04** - *Idempotency:* replaying a request with the same idempotency key returns the original response without creating a duplicate order. Status: verified.

5.2 Non-functional requirements

- **REQ-05** - p95 latency below 350 ms for order submission under 100 req/s sustained load.
- **REQ-06** - Availability target of 99.9 % per calendar month.
- **REQ-07** - All order payloads logged with correlation ids for audit purposes.

5.3 Acceptance criteria

Criteria	Verification	Result
REQ-01, REQ-02	Load test with 250 line items, p95 < 350 ms	Pass
REQ-03	Stale price list returns 400 with line item reference	Pass
REQ-04	Replayed key returns identical body and status 200	Pass
REQ-05 to REQ-07	Perf harness and audit log review	Pass

6. Risks, mitigations, and outlook

6.1 Risks

The supply chain for the new power module remains the top risk. The four open escalations concern the same supplier and are tracked in ticket group **ESC-2024-09**. Inventory turns are within tolerance of the target and do not require executive action.

6.2 Outlook

The next fiscal year focuses on the two deferred packages, the production ramp of the redesigned power supply, and preparation of the 2025 platform review. A mid-year checkpoint is scheduled for calendar week 26, and the final steering committee meeting of the next year is planned for 2025-12-16.

The engineering planning office will publish the FY2025 work package breakdown by 2025-01-15. Escalation path is unchanged from the prior report. Approved by the engineering director on 2025-01-08, revision B.
